Response to Technical Inspection Comments
Manuscript ID: [TBD]
Adaptive Multi-Objective Chaotic S-Box Design for Lightweight and Secure IoT Communication

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Dear Editor,

Thank you for the thorough technical inspection of our manuscript. We have carefully addressed each comment raised during the inspection. Below, we provide a point-by-point response detailing the changes made to the manuscript. All changes are marked in the revised manuscript file `main_Scientific_Reports.tex` and reflected in the compiled PDF `main_Scientific_Reports.pdf`.

Summary of Revisions

Comment #	**Description**
1	Please provide a Funding Declaration in the manuscript. If there was no Funding, please state so.
2	We request authors to provide the same Data Availability statement from the manuscript and update in the "Submission system".

Table 1: Technical inspection comments addressed in this revision.

Point-by-Point Response

Comment 1: Funding Declaration

Editor Comment

Please provide a Funding Declaration in the manuscript. If there was no Funding, please state so.

Our Response

We confirm that this research did not receive any external funding from any funding agency, grant, or individual. No financial support was obtained from governmental, institutional, or private sources for the conduct of this study, the preparation of the manuscript, or the decision to submit it for publication.
To address this comment, we have added a dedicated **Funding** section to the manuscript, placed between the *Acknowledgements* and *Author information* sections. The section contains the standard Nature Portfolio declaration for unfunded research:

> **Funding**
> This research received no external funding.

Location in the revised manuscript: The Funding declaration appears in the unnumbered *Funding* section after Acknowledgements and before Author information in `main_Scientific_Reports.tex` and the compiled manuscript PDF.
Verification: The declaration reads: "This research received no external funding."

Comment 2: Data Availability Statement in Submission System

Editor Comment

We request authors to provide the same Data Availability statement from the manuscript and update in the "Submission system".

Our Response

The manuscript Data Availability section now reads: "The simulator, analysis scripts, configuration, paired seed outputs, and figures supporting this study are publicly archived in Zenodo version v1.0.0 (DOI: 10.5281/zenodo.23439389) and mirrored in the GitHub repository rawnjith/adaptive-multi-objective-chaotic-sbox. No third-party dataset is used." The corresponding author should select "YES" in the submission-system Data Availability field and paste this statement verbatim so the external field matches the manuscript exactly.

Revision status

The latest manuscript PDF is 11 pages and contains 9 cited references, 7 figures, and 1 table (8 display items). The Funding declaration is present. The Data Availability section cites Zenodo DOI 10.5281/zenodo.23439389 and the public GitHub repository. Before resubmission, the corresponding author should enter the exact manuscript statement in the submission-system Data Availability field and confirm it is set to "YES".

Sincerely,

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(on behalf of all co-authors)
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